RESEARCH PROGRAM · PROCESSOR

The AB Processor: A Level Research Program P0–P6

Research Program Monograph · Version 1.0.0

The AB-cloud ISA, the exact Givens-QR compiler, the "reads, does not carry" principle and nonlocality without signaling: stage P0 is executed, the P0–P6 roadmap is chained with verification.

$1-F \le 4.4e-16$	$\le 7/29$	P0
COMPILATION 4×4	GATE DEPTHS	STAGE DONE

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Research Program: The AB Processor

The processor program turns the mathematics of the core into an executable machine: the AB-cloud ISA, the exact compiler, memory, ALU, zero reading, nonlocality, benchmarks and the passport. Stage P0 is executed; the P0–P6 roadmap is chained with verification.

1. Program goals

- fix the AB-cloud ISA: admissible clouds, parameters, composition;
- compile unitary logic exactly: 1−F at the machine epsilon level;
- implement "reads, does not carry" algebraically (V23);
- provide nonlocality without signaling (V24) and obtain the Datasheet 1.0 passport.

2. Methodology

The ISA is not "yet another gate set": every cloud carries a geometric phase, and compilation accounts for it exactly. All stages have pre-registered thresholds and verdict files. Benchmarks follow the E18 methodology: one class, one criterion, strong baselines. numpy replication is mandatory.

3. Executed: P0 (V20)

Table 3.2. The V20 verdict

Indicator	Value
accuracy 4×4	1−F ≤ 4.4e−16; depth ≤ 7
accuracy 8×8	1−F ≤ 6.7e−16; depth ≤ 29
compilation time	1.8 s for the full run
numpy replication	0 mismatches; |Δ(1−F)| = 2.2e−16
status	PASS; stage P0 closed

4. The P0–P6 roadmap

Table 4.1. Processor stages

Stage	Campaign	Content	Depends on
P0	V20	ISA + exact compiler	Th5 (block structure)
P1	V21	volatile memory	ES models, TS1–TS3
P2	V22	AB-cloud ALU	V20
P3	V23	zero reading	E17, Th6
P4	V24	nonlocality	E13/E14
P5	V25	benchmarks	the E18 methodology
P6	V26	Datasheet 1.0	all verdicts of the P-chain

5. Honest boundaries

The 2q KAK optimum (3 gates) is unreachable in the ISA class: the compiler produces 10, and this is a property of the class, not of the algorithm. The V20 targets are unitary 4×4/8×8; non-unitary operators and larger sizes are outside the campaign. The V23/V24 previews are not full verdicts: the full campaigns are ahead.

6. Key results in detail

6.1. From P0 to the P3/P4 previews

V20 — exact compilation (P0)

The only executed processor campaign so far closes stage P0: the ISA is fixed, the Givens-QR compiler assembles unitary 4×4 and 8×8 from AB clouds with infidelity $1-F \le 4.4e-16$ and $6.7e-16$, depths ≤ 7 and ≤ 29 gates, the full run takes 1.8 s. The numpy replication gives zero counter mismatches and $|\Delta(1-F)| = 2.2e-16$. Accuracy at machine epsilon means: further improvement requires changing the arithmetic, not the algorithm.

V23 preview — zero reading

The P3 stage preview: the processor receives six constants of the trace formula and reproduces 12 of 12 ζ zeros exactly; storage = 0. The blind discipline is preserved: the constants contain no zeros, the output is pinned before validation. A preview is not a full verdict: the full V23 campaign requires the whole E17 methodology at the processor level.

V24 preview — nonlocality

The P4 stage preview: a pair of separated clouds gives CHSH S = 2√2 − 4.4e−16, signaling protection 1000/1000, zero commutators of the measuring operators. The correlations are nonlocal but carry no signal — a structural property of quantum mechanics reproduced by the architecture and required for distributed operations.

E18 link — class B

The external control of the processor compiler: against five baselines at equal ISA — a ×59,000 time speed-up at equal accuracy; the honest boundary — the 2q KAK optimum (3 gates) is unreachable in the class (the compiler produces 10). Class B protects the comparison from looseness and makes the result reproducible.

7. How to reproduce

```
python3 modules/processor_ab/scripts/compute_v20.py      # campaign V20
cd modules/processor_ab && python3 -m pytest tests/ -q   # tests
# web lab: modules/processor_ab/web_lab/index.html (compiler tab)
```

8. Verdicts and data

- V20 verdict: modules/processor_ab/data/v20_verdict.json;
- claim audit: modules/processor_ab/verification/claim_audit/;
- figures: figures_hires/ru|en (600 dpi); workbook: tables/qp_workbook.xlsx;
- rpbridge: ISA, compiler, ALU, reader — modules/processor_ab/rpbridge/.

9. Program risks

The main risk is extending the ISA without re-examining the class: any new primitives must pass the E18 methodology again. The second risk is the temptation to claim the KAK optimum: the class boundary is fixed by a verdict and is not removed without new verdicts. The third — the V23/V24 previews: until the full campaigns, their results do not enter the passport.